

Program - 1

Build a Docker Container from a custom Dockerfile

Project Structure:-

Program-1/

-->Dockerfile

-->requirements.txt

-->app.py

Step 1:Create a Dockerfile and add the content

```
mkdir Program-1
```

```
cd Program-1
```

```
nano Dockerfile
```

```
GNU nano 7.2
FROM python:3.10-slim

ENV FLASK_APP=app.py
ENV FLASK_RUN_HOST=0.0.0.0
ENV FLASK_RUN_PORT=5000

WORKDIR /app

COPY requirements.txt .

RUN pip install --no-cache-dir -r requirements.txt

COPY . .

RUN useradd -m appuser

RUN chown -R appuser:appuser /app

USER appuser

EXPOSE 4000
CMD ["python", "app.py"]
```

Step 2:Create a Python File

```
nano app.py
```

```
GNU nano 7.2
from flask import Flask
app = Flask(__name__)

@app.route('/')
def hello():
    return "Hello from custom Docker container!"

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=4000)
```

Step 3: Create a test file requirements.txt

```
nano requirements.txt
```

```
GNU nano 7.2
flask
```

Step 4: Execute docker build command

sudo docker build -t program-1 .

```
rohit@JD:~/my-docker-app$ docker build -t program-1 .
[+] Building 2.8s (12/12) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 854B
=> [internal] load metadata for docker.io/library/python:3.10-slim
=> [internal] load .dockerignore
=> => transferring context: 28
=> [1/7] FROM docker.io/library/python:3.10-slim@sha256:2a8c7d9efdae9b636de80a187a4ce31374ffb3883a7be06232d25a96e8538bf4
=> [internal] load build context
=> => transferring context: 93B
=> CACHED [2/7] WORKDIR /app
=> CACHED [3/7] COPY requirements.txt .
=> CACHED [4/7] RUN pip install --no-cache-dir -r requirements.txt
=> CACHED [5/7] COPY . .
=> CACHED [6/7] RUN useradd -m appuser
=> CACHED [7/7] RUN chown -R appuser:appuser /app
=> exporting to image
=> => exporting layers
=> => writing image sha256:a8ef4af07c414c90140610b721c9c8bc84a9ce2c7f029660bccc1c02a5356d99
=> => naming to docker.io/library/program-1
```

Step 5:Execute docker run command

sudo docker run -d -p 4000:4000 program-1

```
rohit@JD:~/my-docker-app$ docker run -d -p 4000:4000 program-1
749fdd7b238233661ccd76a2ea83fef161329010815c12de3fb69a6ddd83e637
```

Step 6:Test the container in the browser through localhost

http://localhost:4000



